GUI Testing as a Kimiya Program: Deterministic Scripts, Calibrated Perception, and a Path to Verified Interaction with GUI, Network, and Cloud Software

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We report a concrete instance of a general thesis: an end-to-end test of interactive software—graphical, networked, cloud-backed—is naturally expressed as a *Kimiya program*, a computation whose control flow is deterministic and whose meaning-laden steps are calibrated language-model instruments. We built a testing harness in which mouse, keyboard, vision, closed-loop audio, network fault injection, and production-cloud verification operate together over a desktop application and two concurrent browsers, and applied it to a deployed product, finding ten production bugs. We then observe that the harness had, unknowingly, implemented the Kimiya discipline: its element-locator is a select instrument, its screen-assertions are judge guards under implicit purposes, its on-screen value reads are grounded gen, and its database, cloud-API, image-hash, and audio-waveform oracles are exactly the trusted check kernel that keeps judge error from becoming a false certificate. We make the correspondence precise, transcribe a representative scenario into Kimiya surface syntax, and—using an operating point *measured* from the harness's own run artifacts ($\beta \geq 0.976$, $\alpha \leq 0.16$, per-call locate recall ≥ 0.975)—derive a numeric reliability certificate for a passing test. The correspondence is not perfect: GUI testing stresses exactly the extension Kimiya defers, computation that *mutates an external world under adversarial timing*, and we propose the constructs a world-effecting Kimiya needs (act/observe/settle, world-frame retry, timed cost) together with the proof obligations they carry. We then *implement* the extension in `kimiya-lang`, a zero-dependency interpreter and compiler for Kimiya surface syntax: the transcribed scenario checks, runs, and compiles; the measured datasheet installs as a first-class instrument grade, moving the computed certificate of the same program from $\theta = 0.047$ (prior-grade, with overclaim warnings) to $\theta = 0.86$ (measured); and the proof obligations become check-time rejections—an unguarded irreversible click, a vision retrieval with no named instrument, a sightless judge on a screenshot panel are type errors, not runtime surprises. Finally, we recast the artifact as an executable guideline: a playbook by which an autonomous language-model agent can test arbitrary GUI software under the same discipline. The larger aim is a route to *formal verification of interactive systems* that neither pretends the world is deterministic nor lets the model's judgment go uncalibrated—using exactly as much determinism, and exactly as much language model, as each step requires, with the split made auditable by a Kimiya certificate.

Additional Key Words and Phrases: GUI testing, LLM-as-judge, program logic, calibrated oracles, end-to-end testing, verified interaction, Kimiya

1 Introduction

End-to-end testing of a graphical application is the only test layer that exercises the product as a user does, and the one teams most often abandon. Selector-bound scripts decay with every interface change; cross-platform products need a separate harness per surface; and whole classes of behavior—audio, multi-user synchronization, degraded networks, the live production backend—are declared out of scope because no convenient handle exists. Two recent capabilities reopen the problem. Vision-language models locate interface elements from natural-language descriptions reliably enough for production use, and "computer-use" agents can operate arbitrary interfaces through screenshots and synthetic input. The dominant response is to build *autonomous testing agents* that explore an application and improvise actions. Autonomy, however, is the wrong property for a regression test: an improvising agent does not run the same test twice, cannot be reviewed as an artifact, and its failures resist attribution.

This paper argues, and demonstrates on a real system, a different position. An interactive test wants to be *deterministic in its control flow and calibrated in its perception.* The steps that decide

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what to do—open this view, present these pages, then end the talk—must be a fixed, reviewable script. The steps that require *meaning*—find the button described as "the red End button," judge whether the screen shows a live-presentation state, read the join code off a dialog—are irreducibly semantic and belong to a language model. And crucially, the *verdict* must never rest on perception alone: a screen that merely looks right must not be able to pass a test. This is precisely the shape of a *Kimiya* program [Anonymous 2026]: deterministic structure around a triad of datasheet-carrying instruments—gen (produce), judge (discriminate), select (find)—with a trusted kernel check for what is decidable, and a program logic that transfers reliability across judged guards at *measured* error rates.

We did not set out to instantiate Kimiya. We built a testing harness for SeenSlide—a deployed slide-sharing product with a PyQt5 desktop capture application, a FastAPI/PostgreSQL cloud backend, and a Svelte web viewer with real-time collaboration—because the product needed testing that its unit suite could not provide. The harness drives, with one step vocabulary: a native desktop application including a GTK file dialog; two *different* real browsers concurrently under distinct user accounts; a synthetic fullscreen presenter standing in for a human; an operating-system-level virtual microphone whose audio is verified, after crossing a WebRTC mesh, at the speaker output; a fault-injecting network relay; and a live production cloud. On eighteen scenarios it found ten production bugs, three of them invisible on screen and caught only by non-visual oracles, including a database-pool defect that returned HTTP 500 to *every authenticated user* and a three-bug cascade in which each fix exposed the next.

Only afterward did the correspondence become clear: the harness had implemented the Kimiya discipline without naming it. That correspondence is this paper's subject.

Contributions.

- **The correspondence (§3–§4).** A precise mapping from an interactive testing harness to Kimiya: locator = select, screen-assertion = judge under an implicit purpose, value- read = grounded gen, and the database/cloud/image-hash/audio oracles = the trusted check kernel. We transcribe a representative two-user scenario into Kimiya surface syntax.
- **A measured datasheet (§5).** We calibrate the vision locator from the harness's own run artifacts—58 runs, 604 model calls—obtaining a conservative operating point, and use it to derive a *numeric* reliability certificate for a passing test, in the style of the Kimiya logic. This turns a green run from a boolean into a priced guarantee, and separates the claims a test certifies at kernel-certainty from those it certifies only at the judge's error rate.
- **An extension for effectful, timed interaction (§6).** GUI testing stresses exactly what Kimiya's core defers: computation over an external, mutating, adversarially timed world. We propose the constructs a world-effecting Kimiya needs— act/observe/settle, a world-frame retry rule, and timed cost—and state the proof obligations each introduces.
- **An implementation (§7).** The extension is realized in `kimiya-lang`: screen acts with verified delivery, observations that carry their capture origin, the locator as a select instrument whose datasheet is *installed* rather than assumed, agent-indexed actors, and the disciplines of §6 as static rejection rules. The scenario of §4.3 executes end-to-end and its certificate is computed, not derived by hand.
- **An executable playbook (§8).** We recast the artifact as a guideline an autonomous language-model agent can follow to test arbitrary GUI software under this discipline, so the method generalizes beyond the one system we tested.

- **A verification agenda (§9).** We argue that this decomposition is a viable route to formal verification of interactive, networked, cloud-backed software: use exactly as much determinism and as much language model as each step requires, and make the division auditable by a certificate whose meaning depends only on declarations and datasheets.

We are explicit about status: the guarantees are conditional on the oracle model (judges being calibratable instruments), the datasheet is measured on one system's interface families, and the extension of §6 now carries a reference implementation (§7) whose disciplines are enforced at check time, but its metatheory is not yet mechanized. What is solid is the empirical core—the harness, the ten bugs, the measured operating point—and the demonstration that a real interactive test *is* a Kimiya program, twice over: once found in the harness, and once written down and executed.

2 Kimiya in Brief

We recall only what this paper uses; Anonymous [2026] give the full calculus, denotational semantics, logic, and metatheory.

Kimiya treats a language model as a processor whose basic operations act on meaning. Its instruction core is three *oracles*, each carrying a *datasheet*—an empirically measured operating point:

- $x := \mathrm{gen}\langle S\rangle(p)$ produces an S-schema-valid value from prompt p; it is *untrusted* (validity v_S is syntactic; no semantic property of the output is assumed).
- $\mathrm{judge}\langle k, \tau\rangle\ \psi$ evaluates a semantic predicate ψ by k oracle samples at acceptance threshold τ, with a measured confusion matrix: false-accept rate $\alpha = \Pr[\mathrm{tt} \mid \psi\ \mathrm{false}]$, true-accept $\beta = \Pr[\mathrm{tt} \mid \psi\ \mathrm{true}]$, and intra-panel correlation ρ.
- $x := \mathrm{select}\langle\rho_r\rangle(q, w)$ retrieves from store w with datasheet-stamped recall $\rho_r = \Pr[i \in x \mid i\ \mathrm{relevant}]$.
- $\mathrm{check}\ \varphi$ is the *trusted kernel*: a decidable predicate (schema validation, a test, a proof check) at certainty 1, cost 0.

Control flow is internalized in judged conditionals (**if** judge ψ **then** C_1 **else** C_2) and *verified retry* (retry C **until** J **budget** B: rerun C from the loop's entry state until judgment J passes or the budget is spent, whereupon the program *abstains* to an explicit outcome ✠). Value identity is a *purpose-indexed judged equivalence* $a \sim_\kappa b$: reflexive, symmetric, and deliberately *non-transitive*—a *tolerance relation* of which classical equality is the zero-tolerance special case. A purpose $\kappa = \langle \mathit{domain}, \mathit{preserve}, \mathit{allow_loss}\rangle$ declares what an identification must preserve and what it may lose; "the same" is always "the same for a purpose."

Programs denote maps in a cost-graded subdistribution monad—divergence is honest missing mass, budget exhaustion an explicit observable ✠, so *failure is never silent.* On top sits a distributional Hoare logic with triples $\{P\}\ C\ \{Q\}_{\geq\theta}@B$: from precondition P, program C commits in a Q-state with probability at least θ within budget B. The rules are parameterized by datasheet quantities; the two we use directly are:

$$\frac{(\alpha,\beta)\ \text{from}\ d \quad \{P\wedge\psi\}C_1\{Q\}_{\geq\theta_1} \quad \{P\wedge\neg\psi\}C_2\{Q\}_{\geq\theta_2}}{\{P\}\ \textbf{if}\ \mathrm{judge}\langle k,\tau\rangle\psi\ \textbf{then}\ C_1\ \textbf{else}\ C_2\ \{Q\}_{\geq\min(\beta\theta_1,(1-\alpha)\theta_2)}}\ (\textsc{Judge-If})$$

$$\frac{(\rho_r,\pi)\ \text{from}\ d \quad \forall i.\ \{P\wedge i\in x\}C\{Q(i)\}_{\geq\theta}}{\forall i\in\mathrm{Rel}(q,w).\ \{P\}\ x := \mathrm{select}\langle\rho_r\rangle(q,w); C\ \{Q(i)\}_{\geq\rho_r\theta}}\ (\textsc{Select})$$

(For readability we omit Kimiya's cost terms from both conclusions, and write (α,β) for what the full rules consume as conservative *effective* ends $(\alpha_{\mathrm{eff}},\beta_{\mathrm{eff}})$ of a deployed panel configuration; our judges are single-sample, where the two coincide.) Two structural facts matter below. First, a judged guard transfers reliability across an *imperfect observation of a latent truth*—the branch genuinely

executes under ψ or $\neg\psi$, but all a run obtains is the judge's record. Second, a claim proved through check carries certainty 1; a claim proved through judge carries only β (or $1 - \alpha$). Kimiya keeps these two grades distinct, and that distinction is the hinge of our analysis.

3 The Harness

We summarize the artifact; the correspondence to Kimiya is §4.

Split of concerns. Scenarios are prewritten step lists (a YAML DSL). A *runner* interprets them; a *locator* wraps a mid-tier vision model (Claude Opus 4.8, invoked headlessly) behind three primitives—locate (natural-language description → bounding box and click point), judge ("does the screen show X?" → boolean with a one-sentence reason), and read ("what value is displayed here?" → a string). Locate results are cached by (screen size, description); a replay mode reruns whole scenarios from the cache with zero model calls. The vision model appears in exactly these three primitives and nowhere else; it never chooses the next step.

The dimensions. Beyond click and type, the harness supplies what interactive testing usually excludes. A *presenter* renders PDF pages fullscreen on a chosen monitor under a schedule, standing in for the human who presents. A *virtual microphone* (a PipeWire loopback whose playback side is a device-class source) becomes the system default input; text-to-speech (piper) speaks one line per slide, and on the output side the harness records the default sink's monitor and asserts waveform peaks—including an inverted mode that proves *silence*. In the WebRTC voice test the speech feeds only the loopback microphone, so any energy at the speaker sink can only have crossed the mesh from the sending browser. A *relay* routes the application's cloud traffic through a local HTTP proxy; killing it is a clean network outage. Two *browser actors* bind names to real, user-prepared Edge and Firefox windows by title; steps carrying a target crop the screenshot to that window and offset clicks into it, so two viewers of the same page cannot be confused.

Oracles. Every scenario ends in checks that are not visual: structural queries against the application's SQLite; perceptual hashing of each stored slide against the source PDF pages (best-match, with an order constraint, because deduplication legitimately merges near-identical pages); the *public* cloud endpoint—what the audience's viewer actually renders—for slide and viewer counts; audio-waveform peaks; and regex evidence in the application's own logs. Artifacts of every run—a film-strip HTML report with the located box and click point on each screenshot, the application log, a consistent copy of its database, and recorded audio—are retained.

Yield. Across eighteen scenarios the harness found ten production bugs, all fixed and shipped, summarized in Table 1. Every one is an *integration* bug beyond the reach of the project's 65-test unit suite; three were invisible on screen and caught purely by oracles; and bugs 8–10 form a cascade in which each fix exposed the next.

4 GUI Testing as a Kimiya Program

We now make the correspondence exact. The claim is not analogy but identity of structure: the harness is an informal, uncalibrated Kimiya interpreter, and its scenarios are Kimiya programs whose datasheets were left implicit.

4.1 The mapping

Table 2 is the core observation. Three entries deserve comment.

Assertions are judgments under a purpose. A harness assertion reads "the live presentation view with a LIVE badge and an End Presentation button." This is a rubric predicate $\phi_\kappa(\text{screen})$ whose purpose κ is implicit in the wording: it *preserves* the named widgets and key text, and *allows loss* of

Table 1. The ten production bugs found by the harness. "Oracle" = caught by a non-visual check; the rest surfaced through scripted interaction plus a judged or read step.

#	Defect (one line)	Caught by
1	config key never read ⇒ all slide data in volatile `/tmp`	DB oracle
2	slide-marker wiring only in the cloud branch of voice recording	audio + log oracle
3	hardcoded white input under a dark-theme white text token	screen judge
4	slide gate armed too late; exempted the app's own frames	cloud oracle + report
5	session slide counter incremented, never decremented on delete	cloud oracle
6	session row never persisted; Sessions view showed "no talks"	scripted flow
7	viewer-count had zero callers (viewers poll; no socket to count)	cloud oracle
8	four backend modules used a never-connected DB pool	HTTP probe
9	`db.transaction()` called but unimplemented; non-atomic writes	exposed by #8
10	welcome bonus granted twice once #9 was fixed	exposed by #9

Table 2. Harness constructs are Kimiya constructs. The right column names the datasheet quantity each carries.

Harness	Kimiya	Carries
locate(desc, screen)	select$\langle\rho_r\rangle$(desc, screen)	recall ρ_r
assert_screen(desc)	judge$\langle k, \tau\rangle$ ϕ_κ(screen)	(α, β, ρ)
read_text(desc)⟶ $v	gen$\langle Str\rangle$(screen), grounded downstream	validity ν
`verify_db` / `verify_cloud`	check φ (trusted kernel)	certainty 1
`verify_slides_match_pdf`	check over a tolerance relation $\sim_{\kappa_{\text{img}}}$	certainty 1
`verify_audio_out`	check φ (peak/silence predicate)	certainty 1
`click` retries, `optional:`, re-run-once	retry · **until** · **budget** ·	—
scenario FAIL + artifacts	abstain / ✠ (explicit, recorded)	—
scenario PASS	commit of a triple $\{P\} \cdot \{Q\}_{\geq\theta}$	—
model calls per scenario	cost in the budget monoid	—

theme, cursor, clock, and unrelated windows. The harness's very robustness—the same assertion passing in light and dark theme, at different window sizes, with the operator's terminal visible beside the target—is exactly what a Kimiya purpose licenses, and what a pixel-diff oracle forbids. Making κ explicit (§4.3) is the discipline the harness was missing.

Perceptual slide matching is a tolerance relation. The oracle `verify_slides_match_pdf` accepts a stored slide s as a capture of source page p when their perceptual hashes lie within a distance threshold, $s \sim_{\kappa_{\text{img}}} p$. This relation is reflexive, symmetric, and *non-transitive*: two frames each within distance 16 of a shared middle can be 24 apart. The oracle's design— best-match against all pages plus a presentation-order constraint, rather than positional $k \leftrightarrow k$ matching—is precisely a refusal to chain the tolerance relation, the operational form of Kimiya's non-transitivity axiom. We did not know we were obeying the axiom; the axiom is what makes the oracle correct.

Reads are untrusted generation, trusted downstream. When the harness reads an 8-character group join code off browser A's success dialog into a variable and types it into browser B, that read is a gen: untrusted. We never relied on it. Its correctness was established *downstream*, by B actually joining and a check against the database membership count—exactly Kimiya's "untrusted generator proposes, trusted kernel disposes."

4.2 Why the harness almost never issued a false certificate

The most consequential structural fact is why, across roughly thirty runs, the vision layer produced *zero* false passes. In Kimiya terms, scenarios *end in* check. A judged guard (assert_screen, locate) gates *progress*, not the verdict. A judge false-accept (α) merely lets the script proceed; the committed claim—`cloud == kept`, per-talk sequence $1..N$, a WAV peak above threshold—is check-graded at certainty 1. So judge error converts into wasted budget and abstention, almost never into a false certificate. This is precisely what Judge-If composed with Check predicts: the reliability that reaches the commit is the product of a judge factor and a kernel factor, and when the kernel factor is 1 on the load-bearing postcondition, the certificate is robust to α. It also explains the observed failure profile: failures were overwhelmingly *correct rejections*—a real bug, real state drift from a prior failed run, or a wrong assertion—rather than silent mispasses. The harness was, without knowing it, exploiting the check/judge grade separation that is Kimiya's central safety property.

A corollary sharpens honesty. Not every harness postcondition is kernel-graded. "The ink stroke renders in browser B" and "the LIVE badge is shown" are judge-graded: certified only at β. A Kimiya reading of a passing run therefore *stratifies* its claims—which hold at certainty 1 (counts, peaks, cloud totals) and which hold only at the judge's operating point—where the current PASS conflates them. That stratification is the paper's practical upgrade to the harness, and it is where an undetected α-error could still hide a product bug.

4.3 A scenario, transcribed

We transcribe the two-user collaboration scenario (create a group in A, carry the join code to B, verify chat both ways) into Kimiya surface syntax, purposes made explicit. act and settle are the world-effecting constructs *this paper proposes* in §6 (read act as "perform this input effect"); they are typeset in **magenta** to distinguish them from **core Kimiya** keywords, which appear in blue. Everything else is core Kimiya, with the purpose of each select noted in a comment exactly as the Kimiya overview does.

```
context k_ui:                        -- locating a described control
  domain     = "GUI control matching a description on a screenshot"
  preserve   = [role, visible_label, affordance]
  allow_loss = [pixel_position, theme, scale, unrelated_windows]

context k_state:                     -- judging an application state
  domain     = "whether a screenshot shows a described UI state"
  preserve   = [key_widgets, key_text, key_status]
  allow_loss = [cursor, clock, theme, unrelated_windows]

-- A creates the group; the join code is generated (untrusted) and read.
act<A> click( select<0.97>("the + button by the groups heading", A.screen) )   -- under k_ui
act<A> click( select<0.97>("the Create Group button", A.screen) )               -- under k_ui
act<A> type ( "W3 Harness Group" )
act<A> click( select<0.97>("the Create submit button", A.screen) )              -- under k_ui
settle<A> until judge<1,1/1> shows(A.screen,
              "a success view with an 8-char join code") under k_state
          within 12s

code := gen<JoinCode>( read("the 8-character join code", A.screen) )   -- untrusted
```

```
-- B joins with the carried code.
act<B> click( select<0.97>("the + button by the groups heading", B.screen) )   -- under k_ui
act<B> click( select<0.97>("the Join Group button", B.screen) )                -- under k_ui
act<B> click( select<0.97>("the Join Code input", B.screen) )                  -- under k_ui
act<B> type ( code );  act<B> key ( Return )
settle<B> until judge<1,1/1> shows(B.screen,
              "the group 'W3 Harness Group' is selected") under k_state
          within 12s

-- chat A -> B, then B -> A, each verified on the receiving screen ...
-- (omitted; identical shape)

-- THE VERDICT: kernel gates, then a judged gate, then commit.
if check( db.group_members("W3 Harness Group") = 2          -- kernel: certainty 1
        /\ db.messages(group) = [m_ab, m_ba] )              -- kernel: certainty 1
then if judge<1,1/1> shows(B.screen, "the chat pane displays m_ab")
        under k_state                                       -- judged: >= beta
     then commit(trace) else abstain
else abstain
```

This listing is no longer only notation. §7 reports an implementation in which a line-for-line rendering of it—actor declarations, indexed acts and observations, the vision select, the shows judgments, both gates—parses, checks, runs, and compiles, committing at $\theta = 0.86$ under the measured datasheet of §5.

Read the final block: it makes explicit what the informal PASS left tacit. A run commits only through two gates. The kernel gate (check on the membership count and message rows) contributes no instrument error: *any* committing run satisfies those conjuncts with certainty. The judged gate ("the chat pane displays m_{ab}") is an instrument reading at the locator's operating point; the certificate carries that conjunct at the datasheet rate, not at 1. A reviewer of the test—or an LLM agent asked whether the collaboration feature is verified—learns from the certificate exactly which guarantees are ironclad and which are probabilistic, and at what rate.

4.4 The reliability of a passing run, numerically

We instantiate the Kimiya logic on a passing scenario using the *measured* datasheet of §5. Consider a scenario with L load-bearing locate steps (each gating progress toward the commit), J judged guards on the path to the commit, and a `commit` whose postcondition is a conjunction of kernel checks φ_{ker} and judged claims ψ_{jud}. Under the harness retry policy (each locate retried, plus a scenario re-run on terminal miss), the per-call locate recall ρ_r composes to an effective $\rho_r^{\text{eff}} \approx 1$ over the campaign (§5); we carry ρ_r symbolically.

The kernel conjunct is certified at 1 by Check. Each judged guard on the path contributes its true-accept factor β (the guard must fire for the run to reach commit; Judge-If on the taken branch). Each judged *postcondition* claim contributes β directly. So the certified reliability of the commit factors as

$$\theta_{\text{commit}} \;\geq\; \underbrace{\left(\rho_r^{\text{eff}}\right)^{L}}_{\text{located}} \cdot \underbrace{\beta^{J}}_{\text{guards fired}} \cdot \underbrace{1}_{\varphi_{\text{ker}}} \cdot \underbrace{\beta^{|\psi_{\text{jud}}|}}_{\text{judged claims}} .$$

For the transcribed scenario, the load-bearing kernel conjuncts (member count, message rows) are the guarantee that matters, and they are certified at 1 regardless of β—this is the formal content

Table 3. Measured operating point of the vision locator, from 58 runs / 604 model calls. Bounds are conservative (Wilson-95 lower for accept rates; rule-of-three upper for unobserved error).

Instrument	Quantity	Value	Basis
judge (assert)	β true-accept	≥ 0.976	158/158; Wilson-95 low
judge (assert)	α false-accept	≤ 0.16	0/19 true-neg; rule of three
select (locate)	ρ_r per call	≥ 0.975	340/343; Wilson-95 low
select (locate)	ρ_r^{eff} after retry	≈ 1.000	0 unrecovered / 58 runs
gen (read)	value error	≤ 1.0	0/3; rule of three (thin)
cost	latency p_{50}/p_{90}	8.9/14.3 s	229 samples
cost	calls / scenario	10.4	604/58

of §4.2. The *judged* conjunct ("B's chat shows m_{ab}") is certified at $\beta \geq 0.976$ (§5). If one instead demanded a single reliability figure for the *whole* commit read conjunctively, with $J = 4$ guards on the path and one judged postcondition, and taking the conservative $\beta = 0.976$ and $\rho_r^{\text{eff}} = 0.999$ over $L = 8$ locates, the composed lower bound is

$$\theta_{\text{commit}} \;\geq\; 0.999^8 \cdot 0.976^4 \cdot 1 \cdot 0.976 \;=\; 0.992 \cdot 0.908 \cdot 0.976 \;\approx\; 0.879,$$

valid with datasheet-confidence $1 - \delta$ (§5 publishes the locator sheet at the campaign's aggregate level). The number is deliberately unglamorous, and that is the point: a green run is not certainty, it is a priced guarantee, and the price is dominated by how many *judged* claims ride on the commit versus how many are discharged to the kernel. The design lesson the arithmetic hands back to the test author is exactly the one §4.2 states: push load-bearing postconditions into check, and the $\beta^{(\cdot)}$ factors stop mattering.

5 A Measured Datasheet for the Vision Locator

A Kimiya instrument is only as trustworthy as its datasheet, and the datasheet must be *measured*, not assumed. We calibrate the harness's vision locator (Claude Opus 4.8 via a headless CLI) from the campaign's own retained artifacts: 58 scenario runs with structured step reports, and the consoles of 55 runs recording per-call latency and attempt-level retry lines. Ground truth is established in two tiers, reported separately: *(T1) outcome-grounding*—a step's latent truth inferred from the run's terminal hard oracles, which are independent of the vision layer; and *(T2) per-event adjudication*—every non-passing perception event in the campaign was root-caused when it occurred (product bug, oracle bug, state drift, or model flake), and that record is reused here. Where zero errors were observed we report the one-sided 95% upper bound (rule of three, $3/n$), never zero.

Table 3 is the result. Three points are worth stating plainly, in the datasheet spirit of admitting what is weak.

The false-accept bound is the honest limitation. We observed no judge false-accept—including a blind re-read of six uniformly sampled accepted assertions, all confirmed correct—but the campaign presented only ≈ 19 genuine true-negative trials to the assertion judge (runs where a described state was truly absent and correctly rejected). The rule of three therefore caps $\alpha \leq 0.16$, no tighter. This is exactly the quantity §4.2 shows the harness is *robust* to (because commits end in check), which is fortunate, because it is the quantity we can least tightly bound from opportunistic data. A purpose- built calibration suite—the standard Kimiya remedy—would drive it down.

Correlation is real and unmeasured. The configuration is single-model, single-prompt; immediate same-input retries are highly correlated (we observed three consecutive misses on one plainly visible element, then success on a later fresh run). We therefore treat an immediate retry burst as *one*

effective sample and rely on the scenario re-run—a genuinely fresh draw—for amplification. This is the uncalibrated shadow of Kimiya's panel construct (`panel [A,B,C] paraphrase_prompts 2`): decorrelation by model family and prompt paraphrase is what would let $k > 1$ actually amplify, and it is the first thing a production version should adopt.

Recall is the certificate-bearing direction. Per-call locate recall is high (≥ 0.975) and, under the retry policy, effectively 1 over the campaign (no unrecovered locate failure caused a false result). This matches Kimiya's asymmetry: a locate *miss* is loud (the step retries or the scenario abstains, visibly), whereas a locate on the wrong element is caught downstream when the subsequent judged/kernel step fails. Recall, not precision, is what the certificate rides on, exactly as the select datasheet demands.

6 Extending Kimiya for Effectful, Timed Interaction

The correspondence of §4 is faithful for the verification layer but strains at the interaction layer, because GUI testing is Kimiya's deferred extension in its most demanding form: computation that *mutates an external world under adversarial timing*. We identify three gaps and propose the constructs and proof obligations a world-effecting Kimiya needs. We present the design here with its metatheory sketched, not mechanized; §7 reports the implementation and what building it fed back into the design.

6.1 Gap 1: the store is an external, mutating world

Kimiya's state is $\Sigma = \mathrm{Var} \rightharpoonup \mathbb{V}$: values under the program's control. The harness's state is a live desktop, two browsers, an audio server, and a production database—an external store $\mathcal{W}$ the program can *act on* and *observe* but not directly read. We propose splitting the interaction primitives:

- act e: perform an input effect on $\mathcal{W}$ (a click at a located point, a keystroke, an audio injection). Its datasheet is a *delivery* contract—probability the effect is delivered, and its cost—not a truth about the resulting state.
- observe$\langle S \rangle$: sample a schema-typed observation of $\mathcal{W}$ (a screenshot, a DB row, a waveform). An observe is the *subject* of a subsequent judge or check; it is where latent world-truth enters as an instrument reading.

The locate-then-click idiom becomes act click(select$\langle \rho_r \rangle$(desc, observe(screen))): observe the screen, retrieve a point on it, act. Crucially, select now retrieves over an *observation*, so its recall datasheet is conditioned on the observation family—which is exactly how the harness's locator behaves (its reliability differs across the PyQt, GTK-dialog, and browser surface families of Table 3).

6.2 Gap 2: retry must reason about world frames

Kimiya's Retry reruns the body *from the loop's entry state*—snapshot semantics—which is sound because the store is program-owned and can be reset. An external world cannot be snapshotted: a click that landed wrong may have opened a dialog or joined a call, and the next attempt inherits it. This is not a corner case; it is the source of the harness's hardest-won engineering (optional steps, leave-then-rejoin, the "dead microphone node" hazard). A world-effecting retry needs a *frame condition* on the body:

$$\frac{\begin{array}{ll}\{P\}\ C\ \{Q\} \text{ on success} & C \text{ is } \mathcal{W}\text{-idempotent up to } \phi_{\mathrm{inv}} \\ (\alpha_{\mathrm{eff}}, \beta_{\mathrm{eff}}) \text{ for } J & \text{on failure } C \text{ runs compensate restoring } \phi_{\mathrm{inv}}\end{array}}{\{P \wedge \phi_{\mathrm{inv}}\}\ \text{retry } C \textbf{ until } J \textbf{ budget } B\ \{Q\}_{\geq (1-(1-g\beta_{\mathrm{eff}})^B)\pi_{\mathrm{acc}}}}\ (\textsc{World-Retry})$$

Here g is the body's datasheet task-family quality (per-round probability of producing a Q-satisfying outcome) and $\pi_{\text{acc}} = g\beta_{\text{eff}}/(g\beta_{\text{eff}} + (1-g)\alpha_{\text{eff}})$ the acceptance precision, both exactly as in Kimiya's Retry rule [Anonymous 2026]; only the premises are new. They are the price of an unresettable store. Either the body is *idempotent up to a declared world-invariant* ϕ_{inv} (re-running a failed click leaves $\mathcal{W}$ within ϕ_{inv} of where it began—the property the harness's `optional:` steps in informally assert), or it carries an explicit compensate action that re-establishes ϕ_{inv} before the next round (the formal version of leave-then-rejoin). ϕ_{inv} itself is a check- or judge-checkable predicate on an observe; when it is judged, its error enters the reliability like any other. The obligation that this premise cannot be met silently—that a body which mutates the world outside ϕ_{inv} is a *type error*, not a runtime surprise—is the interactive analogue of Kimiya's ban on silent semantic equality.

6.3 Gap 3: cost is timed, and time is adversarial

Kimiya's cost monoid prices *calls*; the harness's sharpest methodology rule—never walk playback markers while audio plays, because the locator's seconds of latency are wall-clock drift—has no home in it. The world evolves *while the judge thinks*. We propose a settle **until** J **within** δ construct: repeatedly observe-and-judge until J holds or a real-time deadline δ elapses, with the deadline a first-class cost alongside the call budget. settle is the principled form of the harness's `wait+assert_screen` idiom, and its obligation is a *stability* premise: the judged state must be one the world will hold still for at least the judge's own latency, or the reading is of a state already gone. The impossibility this formalizes—that no amount of judging can certify a claim about a state that does not persist as long as the judgment takes—is the timing analogue of the recall asymmetry: some failures are silent unless the construct forbids them.

6.4 What survives unchanged

The verification layer is untouched by all three gaps. Purposes, the check/judge grade split, budgets, abstention to ✠, and audit locality apply verbatim; the transcribed certificate of §4.3 is already written in the extended surface syntax and its `commit` means exactly what the core logic says. The extension is entirely in *how state is touched*, not in *how reliability is proved*—which is why the harness could obey the verification discipline (and reap its safety) while improvising the interaction discipline (and paying for it in engineering).

6.5 The extended calculus, assembled

Figure 1 assembles the full language: core Kimiya verbatim, plus the four world-effecting productions this paper proposes, set in magenta as everywhere in this paper. Three disciplines are built into the grammar rather than left to convention. First, *the world is only ever touched through* act *and read through* observe: no other production mentions $\mathcal{W}$, so the program-owned store and the external world cannot be silently conflated, and select now ranges over an *observation* (a value in the store), never over the world directly—which is why its recall datasheet is conditioned on the observation family (§5). Second, every extended production carries its own contract: act a *delivery* datasheet (probability the effect lands, and its cost) that asserts nothing about the resulting state; observe a fidelity and cost; settle a real-time deadline δ that is a first-class cost alongside the call budget; and world-frame retry the invariant and compensation obligations of §6.2. Third, a *freshness obligation* ties them together: the observation an act consumes (through the select that produced its target point) must postdate the last effect on the same surface ι—acting on a stale screenshot is a type error, not a runtime surprise. This is the grammatical form of the harness's practice of re-capturing the window immediately before every click, and the timing analogue of Kimiya's ban on silent semantic equality.

surfaces	$\iota ::= \text{desktop} \mid A \mid B \mid \cdots$	declared regions of the world $\mathcal{W}$
effects	$a ::= \text{click}(pt) \mid \text{type}(s) \mid \text{key}(k) \mid \text{audio}(w) \mid \text{net}(up \mid down)$	instance set, extensible
commands	$C ::= \text{skip} \mid x := e \mid x := \text{gen}\langle S\rangle(p) \mid x := \text{select}\langle\rho_r\rangle(q, x')$	
	$\mid C_1; C_2 \mid \textbf{if } J \textbf{ then } C_1 \textbf{ else } C_2 \mid \text{retry } C \textbf{ until } J \textbf{ budget } B$	
	$\mid \text{commit}(e) \mid \text{abstain}$	
	$\mid \text{act}^{\iota}\ a$	delivery datasheet (d_a, c_a)
	$\mid x := \text{observe}^{\iota}\langle S\rangle$	observation of $\mathcal{W}$ at ι
	$\mid \text{settle}^{\iota} \textbf{ until } J \textbf{ within } \delta$	timed observe-and-judge
	$\mid \text{retry } C \textbf{ until } J \textbf{ budget } B \{\textbf{inv } \phi_{\text{inv}}; \textbf{ compensate } C_r\}$	world-frame retry (§6.2)
guards	$J ::= \text{check } \varphi \mid \text{judge}\langle k, \tau\rangle\ \psi$	
predicates	$\psi ::= u \sim_\kappa v \mid u \models_\kappa v \mid u \perp_\kappa v \mid \phi_\kappa(u)$	

Fig. 1. The world-effecting extension, assembled. Black productions are core Kimiya, unchanged; magenta productions are this paper's extension. $\mathcal{W}$ is the external world, touchable only through act and readable only through observe; ι indexes declared surfaces (the harness's browser actors and desktop). Judged predicates ψ now take *observations* as arguments— latent world-truth enters the logic only as an instrument reading of an observe, at datasheet rates.

7 The Extension, Implemented

Between drafting and revision, §6 stopped being only a design. We implemented the world-effecting extension in `kimiya-lang`, a stdlib-only interpreter *and* compiler for Kimiya surface syntax,[1] and the transcription of §4.3 now executes end-to-end. This section reports what the implementation confirms, what it computes that the paper previously derived by hand, and—most usefully—what building it forced the design to say more precisely. The metatheory remains unmechanized: what follows is executable discipline, not proof.

7.1 Coverage, and the one construct still missing

Everything in Figure 1 is implemented. Surfaces are declared as `display` objects (a monitor region, another X server, or another machine over ssh); act$^{\iota}$ is spelled `act<A> screen.click(x, y)` with click, confirm, type, paste (clipboard-backed, for text that per-keystroke synthesis mangles), key, drag, and scroll effects delivered by OS-level input synthesis; observe$^{\iota}$ is `observe screen<A>()`, returning an observation that carries its capture origin, dimensions, and content hash; select over such an observation invokes the vision locator (boxes in image pixels, mapped through the capture origin to absolute coordinates); screen-state judgments use a `shows` relation whose evidence is the observation itself; settle, world-frame retry with `inv`/`compensate`, and the freshness ledger are as specified. The locator instrument runs Claude Opus 4.8, by default through the same headless-CLI path the harness of §3 used—which is what makes installing Table 3 legitimate: the datasheet is a measurement of *this* instrument, not an assertion about a similar one. The last construct of the transcription to land was the grounded screen-read (gen⟨JoinCode⟩ over an observation): since v1.5, screenshots feed gen directly — `gen<Reading>("Read the join code shown", images=[observe screen<A>()])` — as a typed, vision-gated generation whose

[1] https://github.com/phaysaal/kimiya-lang. The interpreter and the compiled runtime are separate code paths kept honest by a smoke suite that runs both and requires identical commitments; the suite also holds one deliberately ill-formed program per discipline rule and asserts each is rejected *with the right diagnosis*.

freshness is re-verified against the observation's content hash before every call, and whose remote routing is disclosed on the certificate as image egress carrying *which seat's* pixels left the machine. With that, every construct in the transcription of §4.3 executes.

The proof obligations of §6 land as *check-time rejection rules*, in the family of Kimiya's ban on silent semantic equality: an unguarded irreversible act, an irreversible act inside a retry body, a world-touching retry with neither invariant nor compensation, a select over an observation with no named instrument or no declared purpose, a judge panel containing a model that cannot see the screenshot it would vote on, and an actor index naming no declared display are all rejected before any model runs, each with a one-line diagnosis.

7.2 The certificate is computed, not derived

§4.4 derived a certificate arithmetic by hand; the implementation computes it. The same program run against prior-grade instrument sheets commits at $\theta = 0.047$ and prints one overclaim warning per locate—the declared recall 0.97 exceeds the prior-grade $\beta \geq 0.60$, and θ takes the measured end, never the programmer's claim. Installing the datasheet of §5 (a one-line import that *refuses* a sheet with no provenance string) moves the same program to:

```
-- certificate ------------------------------------------------
  status : COMMITTED
  value  : "W3 collaboration verified"
  theta  : 0.8599   (locate:k_ui 0.9746 ×4, shows:k_state 0.9763 ×2)
  instrument locate:k_ui : α ≤ 0.16  β ≥ 0.97
      [measured: seenslide harness, 58 runs / 343 locate calls]
  screen : 8 act(s), 4 locate(s)
  actor  : A → :0 (local)     actor : B → :0 (local)
```

Both runs pass; they are not equally strong evidence, and the certificate is the artifact that says so. The stratification that §4.2 proposed as the paper's practical upgrade to the harness is here the default output format.

7.3 What implementation fed back into the design

Four points where executing the design sharpened it.

Delivery contracts need verification, not just a rate. §6 gave act a delivery datasheet—a probability that the effect lands. The first live test on a multi-monitor workstation showed why that is not enough: an L-shaped layout leaves uncovered regions in the X screen's bounding box, and the display server *silently clamps* a pointer sent there to the nearest valid pixel—the click lands at a coordinate the program never named while the trace faithfully records the one it asked for. No delivery *rate* captures this failure, because the effect is delivered, somewhere. The implementation therefore reads the pointer back after every move and refuses divergence; the design lesson is that an act contract should be stated as *verified delivery*—effect confirmed at the named coordinate—rather than delivery probability alone.

Irreversibility is a property of the control, not the action type. The grammar assigns effect classes per action, but whether a click can be undone depends on whether the pixel under it reads "Cancel" or "Delete forever"—which no static class of click can know. The implementation splits the effect: `click` is recoverable, `confirm` is the same input event declared irreversible, and the program states which one it is performing. The verified-gate obligation then attaches to `confirm`. This converts an unknowable property of the world into a reviewable claim in the source—the same move Kimiya makes with declared purposes.

Cached perception has two epistemic grades. The harness cached locator coordinates and replayed them for zero-cost reruns (§3); the implementation had to decide what that *means*. It distinguishes an *exact* hit—the current observation's hash equals the cached one, so the cached boxes are a reading of this very image, re-entering θ at the datasheet rate—from a *replay*—pixels have changed and the program asserts layout stability, an assumption the certificate counts and discloses. Replay is sound for exactly the reason §4.2 identifies: the locate gates progress, never the verdict, so a stale coordinate produces a wrong click, the world diverges, and the never-cached kernel and judge gates refuse to commit. The harness's cache was correct; the implementation says why, and prices it.

Observations are their own disclosure class. A screenshot shipped to a remote model is not a prompt the program composed—it is whatever happened to be on the user's display. The implementation announces, before any model runs, both the input effects it will synthesize and the fact that captured screens will leave the machine, alongside the standard egress banner. A related consequence of Kimiya's cross-provenance rule surfaced untouched: two Opus voters on a `shows` panel are one model family—one pair of eyes certifying its own reading—so the panel runs but is flagged uncertified, forcing a second vision family into any certifying screen judgment.

7.4 Actors, and what independence means

The agent-indexed surfaces ι of Figure 1 are implemented as declared displays with per-seat coordinate spaces and per-seat freshness worlds: a box found in B's capture can only produce a click delivered to B's seat, and acting on A never stales B's observations. Implementation forced one honest distinction the grammar glosses: two actors that resolve to the same X server share one pointer and one keyboard focus, so their indices label *intent*, not independence—the two-monitor staging of the original harness is in this class. Actors on distinct servers are genuinely independent, which we verified live: with a nested X server as seat B, clicking and typing on B left seat A's pointer untouched, and each seat's captures carried its own dimensions, origin, and hash. The certificate prints where every actor resolved, so a reviewer can tell which class a run was in.

8 A Playbook for an Autonomous Testing Agent

The method is not tied to SeenSlide. We state it as a procedure an autonomous language-model agent can follow to test arbitrary GUI software under the same discipline. It is deliberately imperative and checkable; each step names the Kimiya construct it produces.

(1) **Fix the control flow first; never improvise it.** Write the scenario as a deterministic step list (a user story: open X, do Y, then Z). The agent may *author* the script using its judgment, but the *run* must replay it exactly. A step list is a reviewable artifact; an exploration trace is not.

(2) **Make every meaning-laden step a named instrument.** Locating a control is select (a description + a recall expectation); judging a state is judge (a description + a purpose); reading a value is gen (untrusted until grounded). Do not let the model choose the next *action*—only perceive.

(3) **Declare a purpose for every judgment.** State, per assertion, what must be preserved (the widgets/text that make the state that state) and what may be lost (theme, position, clock, unrelated windows). This is what makes the assertion robust and well-posed; an assertion without a purpose is a pixel-diff in disguise.

(4) **End every scenario in a non-visual kernel check.** A test that concludes on a screen judgment certifies nothing at certainty 1. Find the ground truth the feature is *about*—a database row, a public API response, a file on disk, an audio waveform, a log line—and assert

it with check. This is the single most important rule: it is what makes judge error convert to abstention rather than a false pass (§4.2).

(5) **Stratify the certificate.** In the `commit`, separate kernel-graded conjuncts (certainty 1) from judged conjuncts ($\geq \beta$). Report both. A consumer of the test—human or agent—must know which guarantees are ironclad.

(6) **Retry with a declared world-invariant.** Wrap flaky interaction in `retry`/`optional`, but state what re-running is allowed to disturb (§6): either the action is idempotent up to ϕ_{inv}, or it carries a compensation. Treat a call surviving from a prior run (a dialog, a joined call, a dead device handle) as inherited world state, not a clean slate.

(7) **Wait with** `settle`**, not a fixed sleep, and never judge a transient.** Poll-observe-judge until the state holds or a real-time deadline passes. Do not perceive during continuous change (playback, animation, streaming); the perception latency will read a state already gone.

(8) **Isolate and neutralize side effects.** Run the app under a disposable environment (a fresh home directory, a throwaway identity, a null credential store). For cloud tests, ensure the test identity cannot touch real user data, and purge test artifacts with a filter that names their test origin *in the deletion itself*—never a list gathered from logs. Route network through an injectable proxy so outages are a test action, not a host change.

(9) **Measure your instruments before you trust them.** Retain every run's screenshots, judgments, and outcomes. Periodically re-adjudicate a sample against the terminal kernel oracles to estimate (α, β, ρ_r) and latency; publish the datasheet; recompute the scenario's certified reliability (§4.4). Report unobserved error as a one-sided bound, never as zero.

(10) **Decorrelate to amplify.** A single model retried on the same prompt barely amplifies. When a judgment is load-bearing and cannot be pushed to the kernel, vote across model families and prompt paraphrases (a panel), and treat an immediate same-input retry burst as one effective sample.

Steps 4 and 5 are the discipline's core; a large-language-model agent that follows only those two already tests far more safely than an autonomous explorer, because its verdict stops depending on its own eyes. The remaining steps buy tighter numbers and safer interaction.

9 Toward Verified Interactive Systems

The wider motivation is a route to *formal verification of interactive, networked, cloud-backed software*—systems that have long resisted it for two reasons this decomposition addresses head-on.

Such systems resist verification because parts of their specification are *semantic*—"the viewer shows the presenter's current slide," "the audience hears the speaker," "the schedule was imported correctly"—and no decidable predicate captures them; and because their state is an *external world* no proof assistant can model in full. The usual responses each fail one half: classical model checking demands the whole system reduce to a finite transition system, which the semantics and the world both refuse; and pure LLM-agent testing certifies nothing, because an improvising agent's judgment is uncalibrated and its runs are not artifacts.

A Kimiya program is a third option, and the harness is its existence proof. It uses *exactly as much determinism as the task requires*— the control flow, the kernel oracles, the world-invariants—and *exactly as much language model as the task requires*—the locates, the state judgments, the grounded reads—and the boundary between them is not a matter of taste but of what each step *can* be: decidable predicates go to check, meaning-laden ones to the instruments. The certificate then states precisely what was verified at certainty 1, what was verified at a measured error rate, and what was

left uncovered—and, by audit locality, its meaning depends only on the program's declarations and the cited datasheets, so a reviewer (or a downstream agent) can trust a test without replaying it.

Concretely, this suggests a research program. *(i)* A world-effecting Kimiya core (§6) with a mechanized soundness theorem for World-Retry and settle, reducing the external world to an oracle-observed store with declared invariants. *(ii)* A calibration methodology for interface-perception instruments —purpose-built suites per surface family (native toolkit, web, mobile), published as datasheets an agent can cite. *(iii)* A compiler from user-story specifications to Kimiya test programs, closing the loop the playbook opens by hand. The end state is a testing regime in which a green run is not a boolean but a signed certificate—"this feature holds at certainty 1 on these kernel facts and at $\geq \beta$ on these judged facts, within this budget, citing these datasheets"—for exactly the software that today has no verification story at all.

10 Related Work

LLM-driven GUI testing is predominantly mobile and autonomous: scenario-guided and multi-agent exploration *generate* actions, where we execute fixed scripts and delegate only perception; desktop is a recognized gap. *Commercial AI test tools* offer natural-language authoring and self-healing selectors, philosophically akin to our select, but scoped to web/mobile and silent on audio, native dialogs, and cross-application flows. *Multi-user feature testing* coordinates LLM agents on separate mobile devices; we test two-user collaboration deterministically, with the model reading a runtime-generated code across browsers. *Voice/WebRTC testing* evaluates STT/LLM/TTS pipelines, typically via fake media streams; we found no system closing the audio loop at the operating-system level inside a general GUI harness. Foundationally, we build on Kimiya [Anonymous 2026]: our contribution is to show that an independently constructed interactive test *is* a Kimiya program, to measure a real instrument datasheet, and to identify the world-effecting extension the domain forces. (A full bibliography accompanies the camera-ready; the companion technical report enumerates the harness's ten bugs and scenarios in full.)

11 Limitations

The guarantees are conditional on the oracle model—judges being calibratable instruments—which our datasheet supports in a restricted regime (short, grounded, discriminative visual judgments) and no further. The datasheet is measured on one product's interface families and one model; the α bound is loose for want of true-negative trials (§5), and correlation is treated conservatively rather than measured. The world-effecting extension (§6) now has a working implementation (§7) whose disciplines are enforced as check-time rejections, but its metatheory remains sketched, not mechanized, and the implementation's own guarantees are tested, not proved. Its remote (ssh) seats are exercised only to the point of command construction and failure handling, not against a second machine; and the grounded screen-read, though implemented, has been exercised against the mock oracle and local models only — no measured datasheet exists yet for read accuracy, so a certificate carries it at prior grade. The harness is Linux/X11-specific in its input and capture layers, requires exclusive use of the screen, and cannot test sub-second interactive behavior because perception latency is seconds. Finally, our evidence is one system and one operator: the ten bugs are an existence proof of effectiveness, not a controlled comparison. What we claim is narrow and, we believe, solid: that a real end-to-end interactive test, built without knowledge of the theory, turned out to be a Kimiya program, and that reading it as one both explains its safety and shows how to make it a certificate.

12 Conclusion

We set out to test a real interactive product and, in doing so, built a harness that drives desktop and dual-browser surfaces, closed-loop audio, network faults, and a live cloud from one deterministic step vocabulary, finding ten shipped bugs. Reading the result through Kimiya, we found the harness had implemented the theory unaware: its locator is select, its assertions are judge under implicit purposes, its reads are grounded gen, and its database and audio oracles are the check kernel whose presence is precisely why judge error almost never became a false pass. We measured the locator's operating point from the run artifacts, derived a numeric certificate for a passing test, and identified the one place the correspondence strains—the external, timed world—proposing the constructs a world-effecting Kimiya needs, and then implementing them: the transcribed scenario now checks, runs, and compiles, its certificate computed at the measured operating point, its proof obligations enforced as type errors, and its perception cache priced by epistemic grade. The method generalizes to a playbook any language-model agent can follow. We read all of this as a first, concrete step toward verifying the interactive, networked, cloud-backed systems that have no verification story today: use exactly the determinism and exactly the language model each step demands, and let a Kimiya certificate make the division auditable.

References

Anonymous. 2026. Kimiya: A Program Logic for Semantic Computation with Language Models. *Manuscript.*